

```

fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
%fit_data_file = 'FitEn_cut800br200_2D_fit_bgSer.mat';%Externaly 2-exponent fit
fixed bg at high momentum.
%fit_data_file = 'FitEn_cut800br100_2D_dirGH_dE20_fix_bg_slope.mat'

```

```

fit_data_file =
'FitEn_cut800br100_2D_dirGH_dE20_fix_bg_slope.mat'

```

```

fit_data_file = 'FitEn_cut800ref200_2D_dirGH_dE20_fix_bg_slope.mat'
if ~exist('fit_resEi800','var')
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi800')
        fit_resEi800 = ld.fit_resEi800;
    else
        fit_resEi800 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_cut_data_file = 'multicuts_fit_dataGH_ei800meV.mat';
if ~exist('cuts2fit800','var')
    ld = load(fullfile(data_path,fit_cut_data_file));
    cuts2fit800 = ld.cuts2fit800;
end
cuts2fit = cuts2fit800;
fit_res = fit_resEi800;
mi = maps_instrument(787,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
c1 = {cuts2fit800.cut2fit800_off200}

```

```

c1 = 1x1 cell array
{1x1 sqw}

```

```

for i=1:numel(c1)
    c1{i} = c1{i}.set_instrument(mi);
    c1{i} = c1{i}.set_sample(sample);

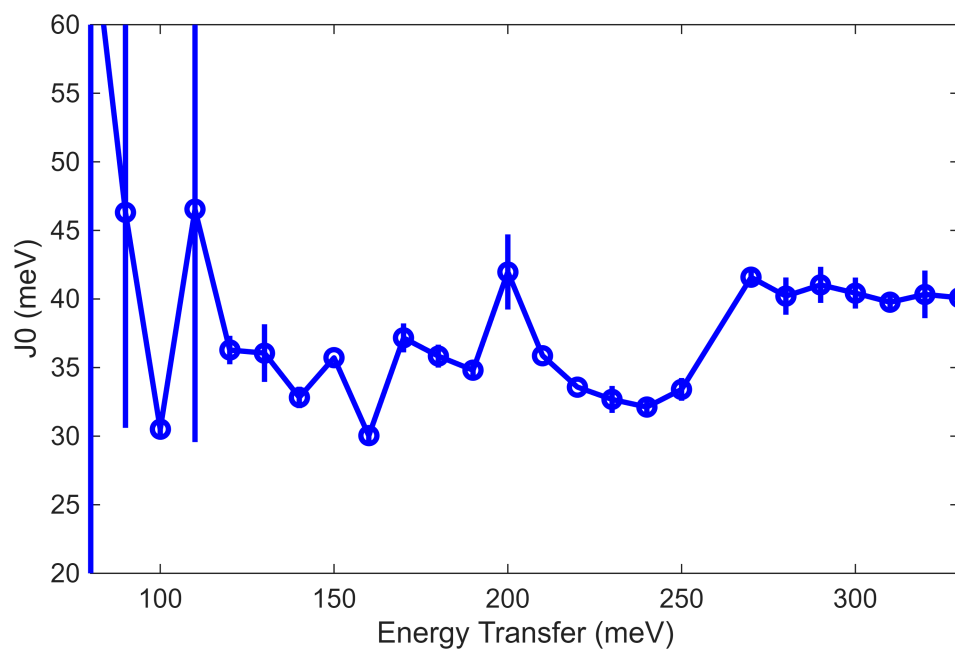
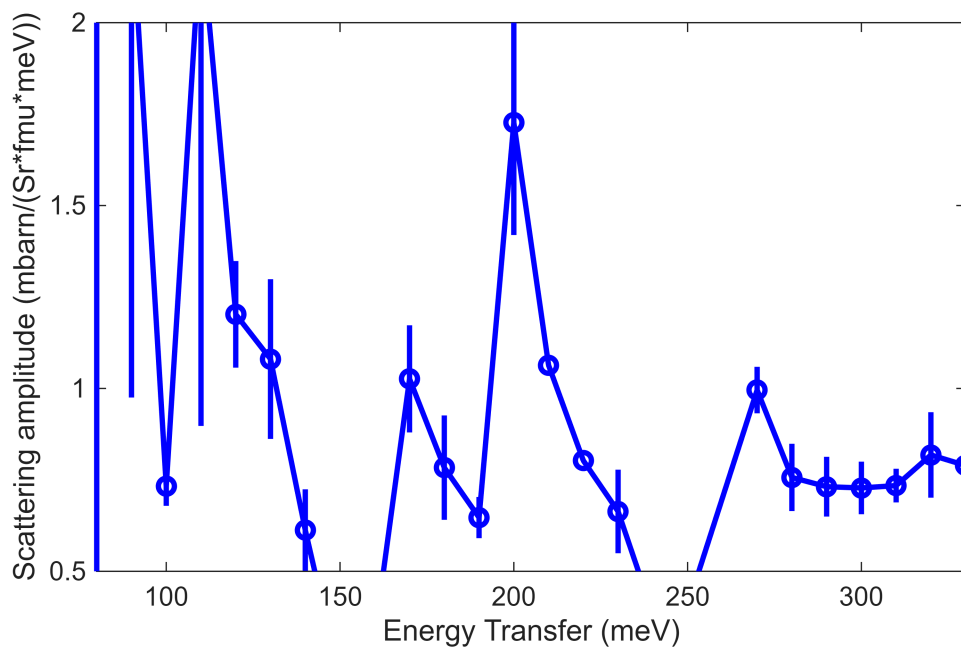
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end
%cuts2fit800.cut2fit010_dir100 = c1{1};
the2Dcuts =c1;

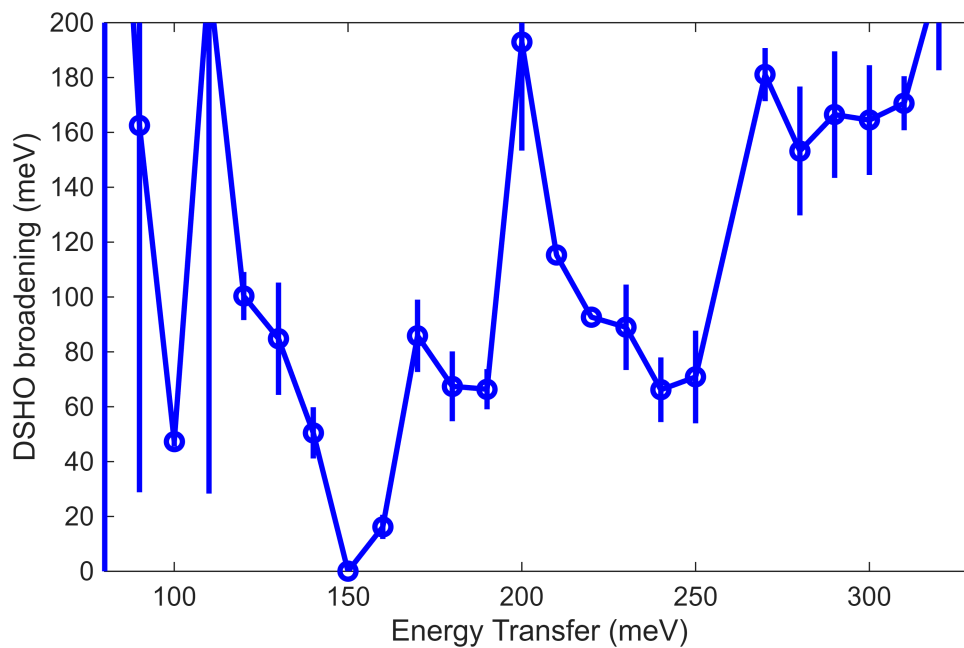
```

```

[Sgn1,J0,G] = plot_SJG_results(fit_res,'b',{ 'S','J0','gamma'},{[0.5,2],[20,60],
[0,200]});

```





```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1))
```

```
dE =  
10
```

```
e_min = Sgn1.x(1);
```

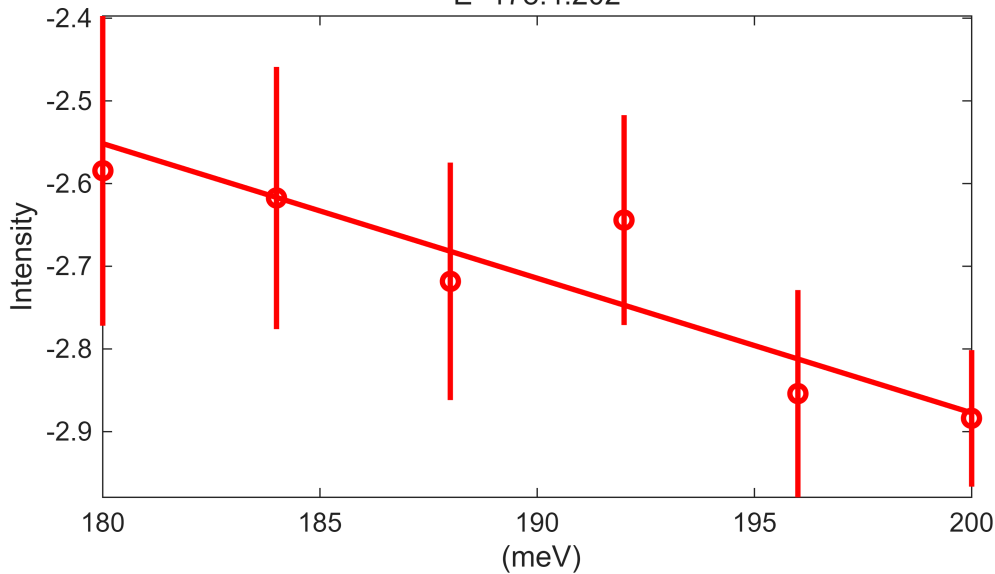
```
En = 200
```

```
En =  
200
```

```
dE_step = 4; %original energy transfer step data were binned to. No point in  
going finer  
half_dE = 10;  
en_idx = floor((En-e_min)/dE)+1;  
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);  
%the2Dcuts =c1;  
  
%init_bg_params =  
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg  
%in case of 2-D fitting  
%  
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p  
arams,init_bg_params,false)  
%the2Dcuts{3} = cut(the2Dcuts{3},[-1,0.02,1],[]);  
lfit_par = fit_res.all_fit_par(en_idx);  
en_range = lfit_par.en_range;  
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2  
,true,false);
```

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16584

Fe_ei800, w2_800off200 Sym fits.sqw
 $2 \leq \zeta \leq 1.6$ in $[2, \zeta, 0]$, $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2,$
 $E=178:4:202$



Warning: Dataset 4:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.7619

Beginning fit (max 20 iterations)-----
Starting point

Total time = 0.80553s Reduced Chi^2 = 3.224

Free parameter values:

66.39 0.647 34.81 0 0 0 0

Iteration = 1

Total time = 3.3036s Reduced Chi^2 = 0.9219 Levenberg-Marquardt = 0.1

Free parameter values:

61.24 0.3389 34.95 -0.02046 -0.0237 -0.02544 -0.0135

Iteration = 2

Total time = 5.4702s Reduced Chi^2 = 0.91882 Levenberg-Marquardt = 0.01

Free parameter values:

28.53 0.2547 33.2 -0.02058 -0.02329 -0.02525 -0.01329

Iteration = 3

Total time = 7.6498s Reduced Chi^2 = 0.91557 Levenberg-Marquardt = 0.001

Free parameter values:

35.65 0.2758 33.88 -0.02054 -0.02353 -0.02527 -0.01337

Iteration = 4

Total time = 9.7787s Reduced Chi^2 = 0.9162 Levenberg-Marquardt = 0.0001

Free parameter values:

29.04 0.2468 33.76 -0.01969 -0.0223 -0.024 -0.01276

Total time = 10.3212s Reduced Chi^2 = 0.91237 Levenberg-Marquardt = 0.001

Free parameter values:

29.24 0.2478 33.77 -0.01972 -0.02234 -0.02405 -0.01278

Iteration = 5

Total time = 12.4509s Reduced Chi^2 = 0.91814 Levenberg-Marquardt = 0.0001

Free parameter values:

19.69 0.2111 33.49 -0.01864 -0.02094 -0.02225 -0.01214

Total time = 12.9781s Reduced Chi^2 = 0.91471 Levenberg-Marquardt = 0.001

Free parameter values:

19.89 0.2121 33.49 -0.01868 -0.02098 -0.0223 -0.01216


```

Total time = 13.5098s    Reduced Chi^2 = 0.91666    Levenberg-Marquardt = 0.01
Free parameter values:
    21.32    0.2188    33.52    -0.0189    -0.02127    -0.02263    -0.0123
Total time = 14.0486s    Reduced Chi^2 = 0.91839    Levenberg-Marquardt = 1
Free parameter values:
    27.12    0.2437    33.68    -0.01964    -0.02227    -0.02382    -0.01276
Total time = 14.5733s    Reduced Chi^2 = 0.92069    Levenberg-Marquardt = 100
Free parameter values:
    28.98    0.2474    33.76    -0.01971    -0.02233    -0.02402    -0.01278
Total time = 15.115s    Reduced Chi^2 = 0.91668    Levenberg-Marquardt = 10000
Free parameter values:
    29.21    0.2477    33.77    -0.01972    -0.02234    -0.02405    -0.01278
*** No improvement in chi-squared over previous iteration ***
Total time = 15.1231s    Reduced Chi^2 = 0.91237
Free parameter values:
    29.24    0.2478    33.77    -0.01972    -0.02234    -0.02405    -0.01278

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	29.24	+/-	6.308
4	0.2478	+/-	0.01793
5	0		
6	33.77	+/-	1.14
7	0		
8	0		
9	0		
10	0		

Background parameter values:

Function 1:

1	0.4806		
2	-0.009889		
3	-0.01972	+/-	0.001997

Function 2:

1	0.4094		
2	-0.007553		
3	-0.02234	+/-	0.002145

Function 3:

1	1.285		
2	-0.01337		
3	-0.02405	+/-	0.002367

Function 4:

1	1.461		
2	-0.01628		
3	-0.01278	+/-	0.002647

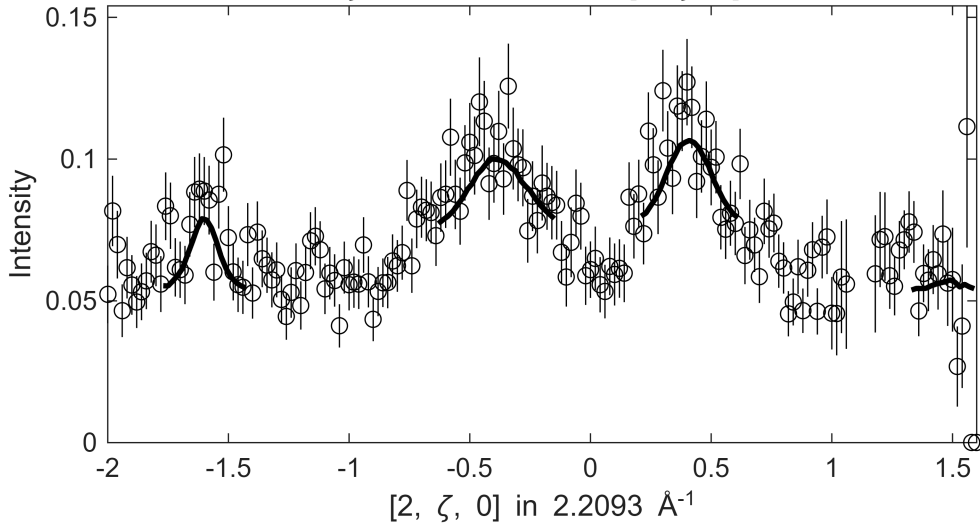
Covariance matrix for free parameters:

1.0000	-0.0658	0.1325	0.3818	0.4076	0.4088	0.4700
-0.0658	1.0000	0.1994	-0.0887	-0.2317	-0.2481	0.5817
0.1325	0.1994	1.0000	0.1078	0.0941	0.1243	0.2151
0.3818	-0.0887	0.1078	1.0000	0.0644	0.0735	-0.1693
0.4076	-0.2317	0.0941	0.0644	1.0000	0.1245	-0.1954
0.4088	-0.2481	0.1243	0.0735	0.1245	1.0000	-0.2092
0.4700	0.5817	0.2151	-0.1693	-0.1954	-0.2092	1.0000

Fe_ei800, w2_800off200 Sym fits.sqw

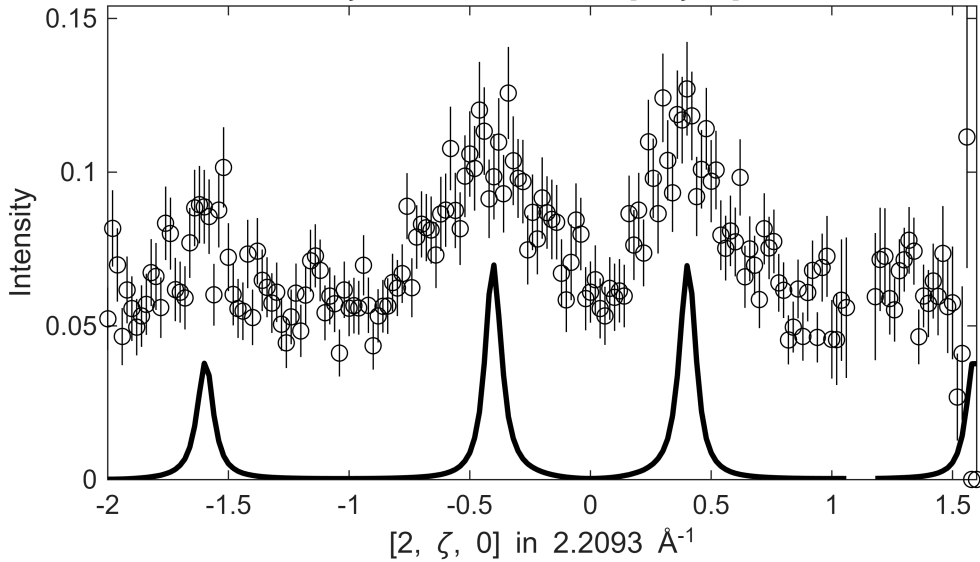
$S=0.25\pm0.018$; $J_0= 34\pm 1.1$; $\gamma=29.2405\pm6.30758$

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $180 \leq E \leq 200$
 $\zeta=-2.01:0.02:2.01$ in $[2, \zeta, 0]$



Fe_ei800, w2_800off200 Sym fits.sqw

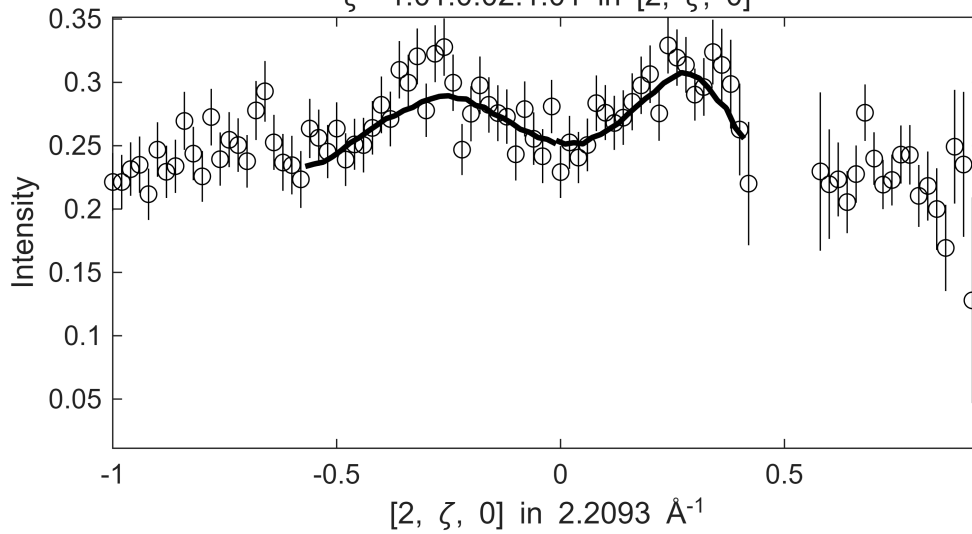
$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $180 \leq E \leq 200$
 $\zeta=-2.01:0.02:2.01$ in $[2, \zeta, 0]$



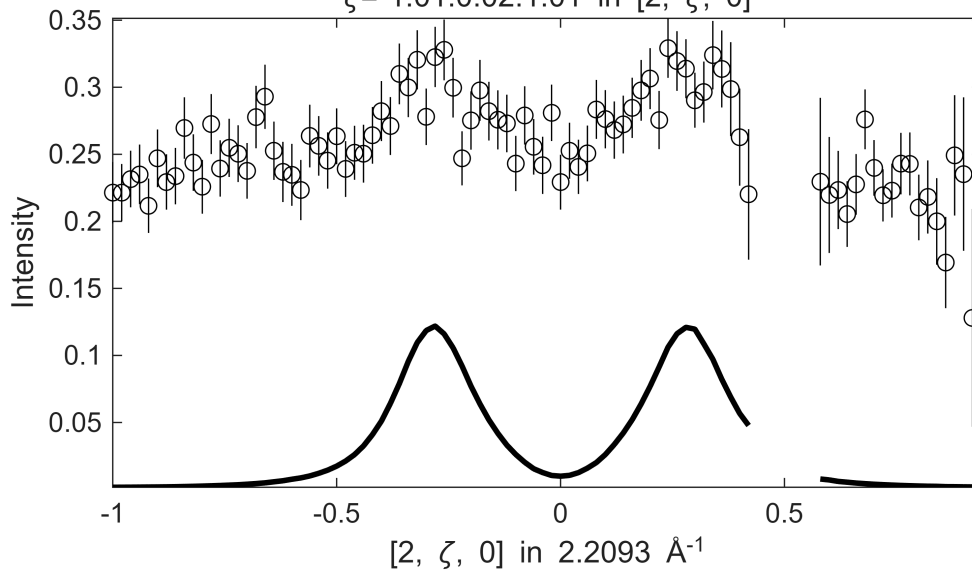
```
%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
Sgn1.signal(en_idx) = fit_par.p(4);
J0.signal(en_idx) = fit_par.p(6);
G.signal(en_idx) = fit_par.p(3);
plot_SJG_results(fit_res, 'b', {'S', 'J0', 'gamma'}, {[0,1.5],[20,60],[0,100]},
{Sgn1,J0,G});
```

```
the2Dcuts{1} = cut(the2Dcuts{1},[-1,0.02,1],[,]);
```


Fe_ei800, w2_800off200 Sym fits.sqw
 $S = 1.4 \pm 0.047$; $J_0 = 42 \pm 3.1$; $\gamma = 135.325 \pm 22.9892$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$



Fe_ei800, w2_800off200 Sym fits.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$



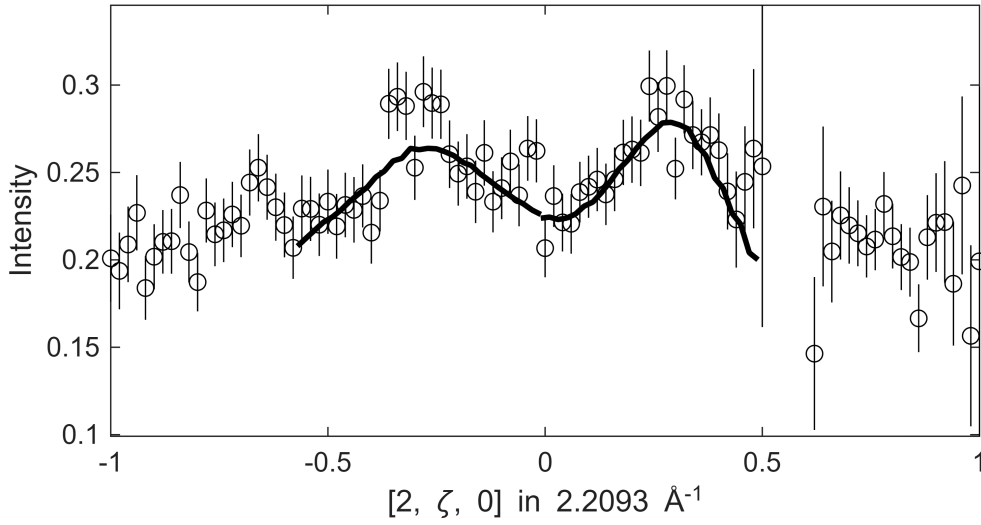
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*****
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*****
*****
*** step: 3#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.16502

Fe_ei800, w2_800off200 Sym fits.sqw

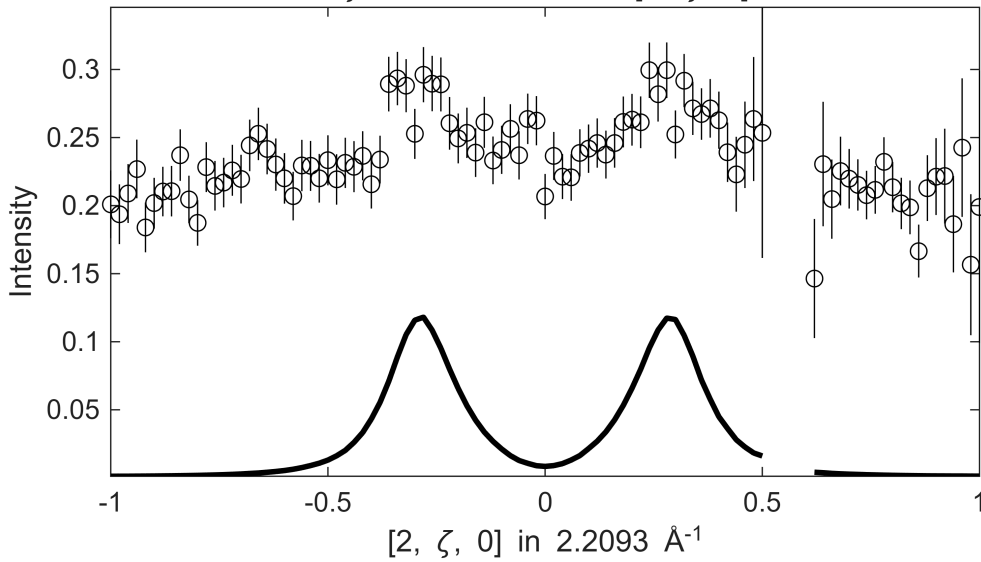
S= 1.1±0.037; J0= 39± 2.5; gamma=90.4158±15.2016

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 90 ≤ E ≤ 110
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



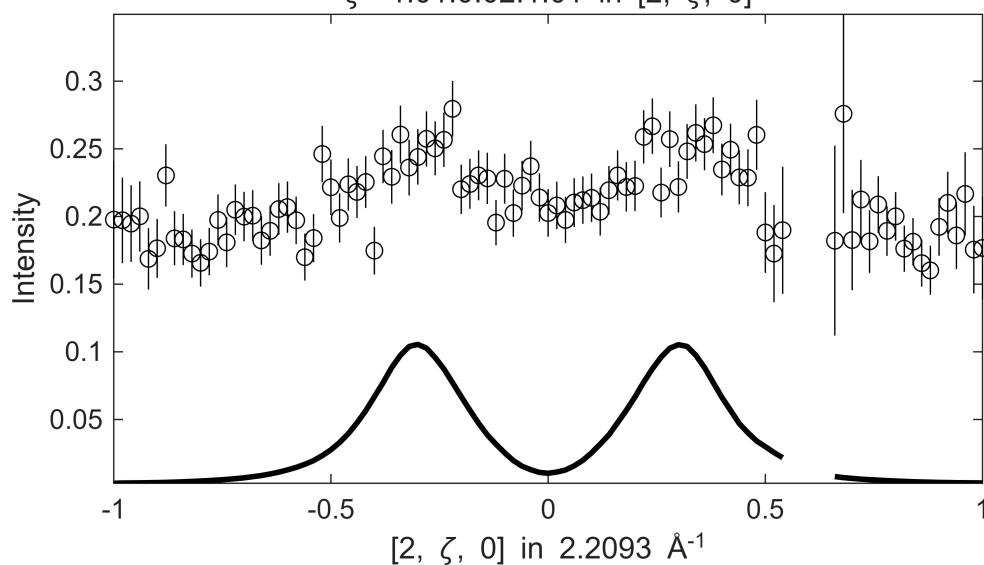
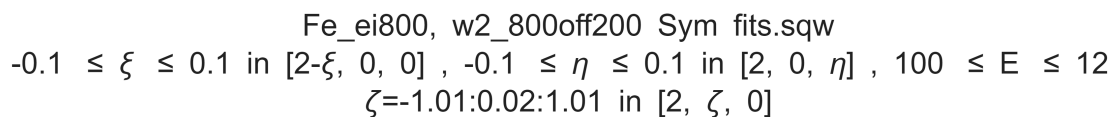
Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 90 ≤ E ≤ 110
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

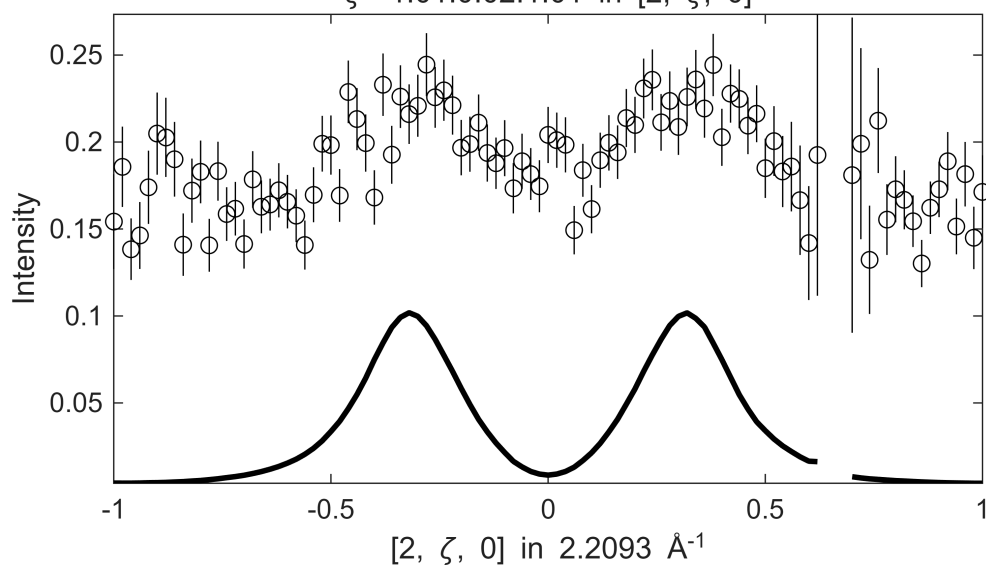
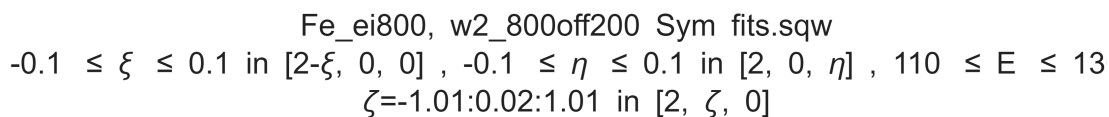


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*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*** step: 4#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 100 ≤ E ≤ 12
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

[illegible]

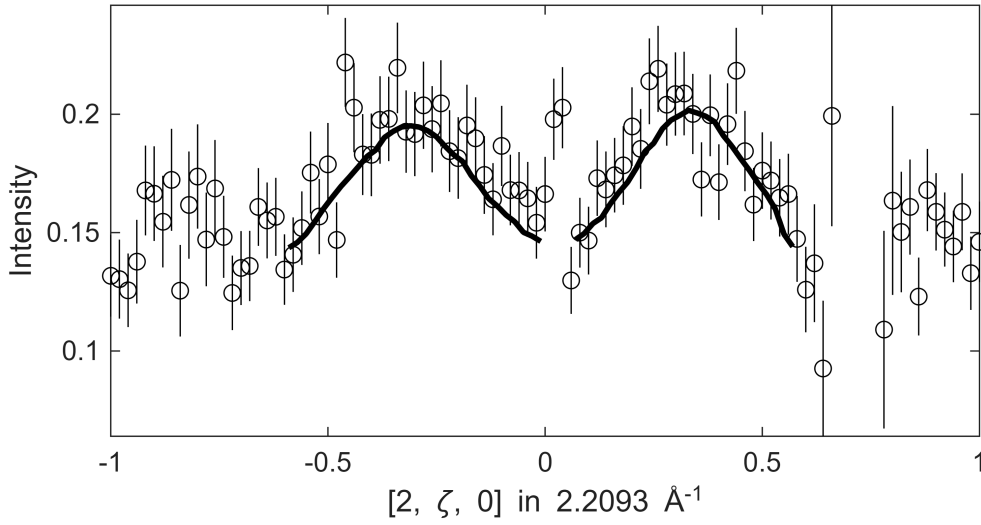
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 110 ≤ E ≤ 13
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

[illegible]

Fe_ei800, w2_800off200 Sym fits.sqw

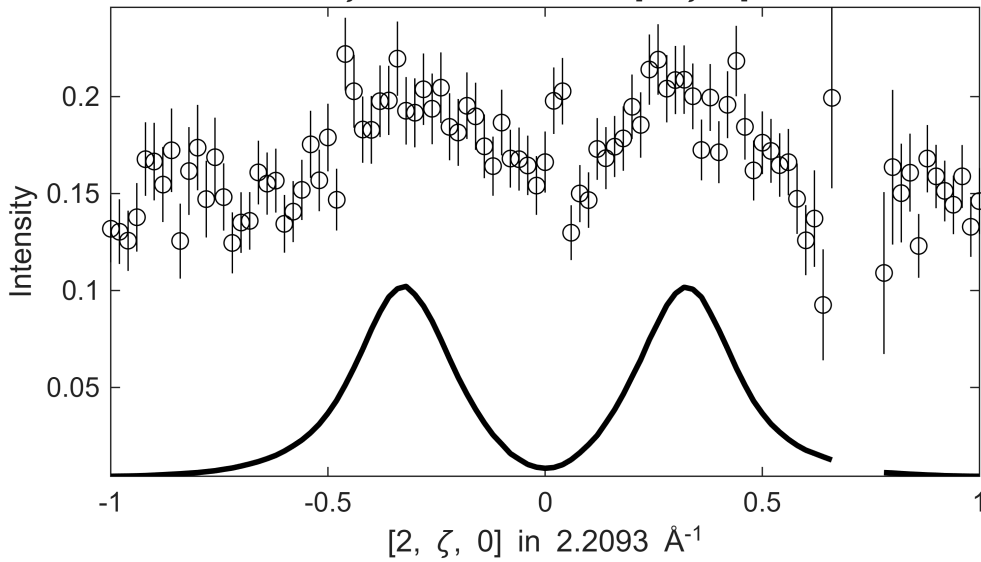
S= 3 ± 0.088 ; J0= 67 ± 4.6 ; gamma= 438.178 ± 58.554

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 14$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$



Fe_ei800, w2_800off200 Sym fits.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 14$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$

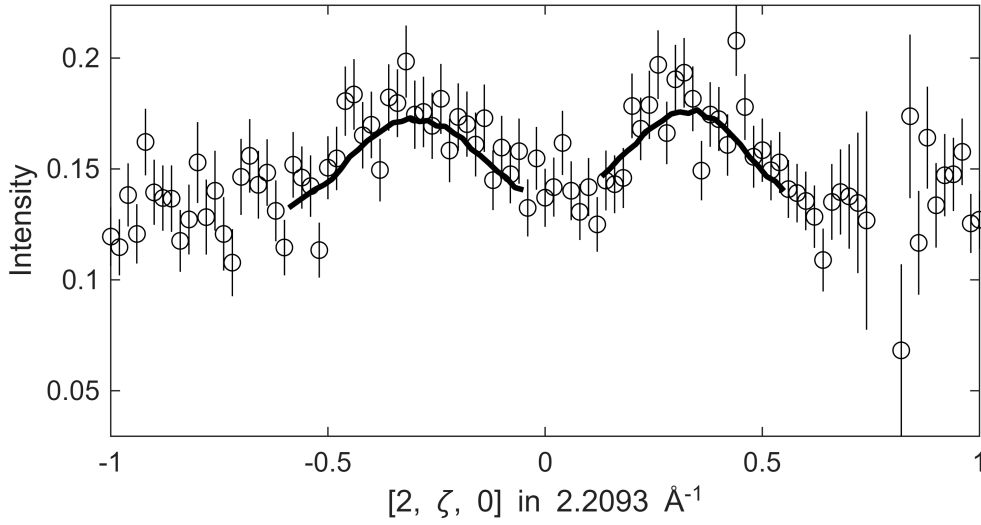


```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*** step: 7#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```


Fe_ei800, w2_800off200 Sym fits.sqw

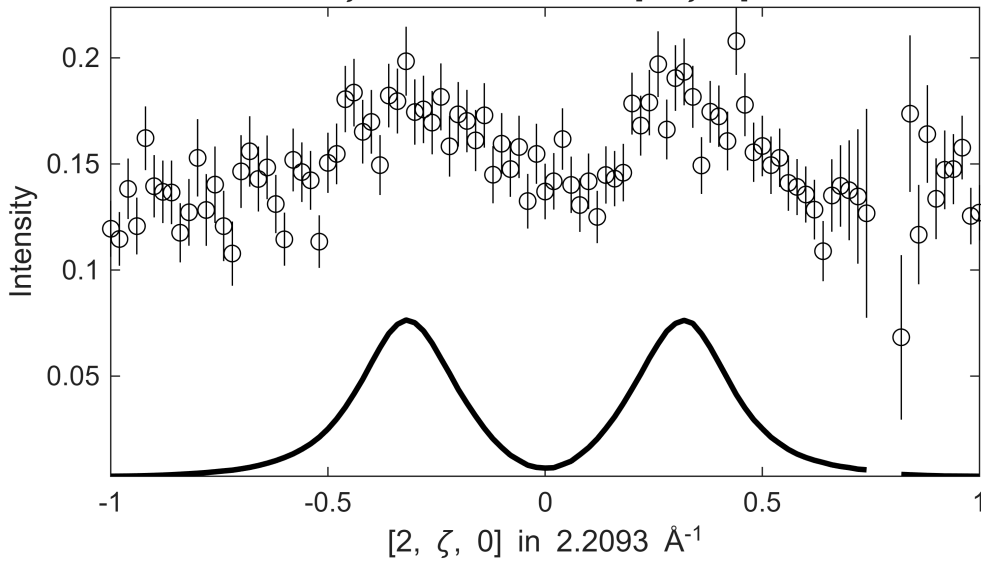
S= 2.6±0.11; J0= 79± 9.1; gamma=528.089±105.623

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 130 ≤ E ≤ 15
 ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 130 ≤ E ≤ 15
 ζ=-1.01:0.02:1.01 in [2, ζ, 0]

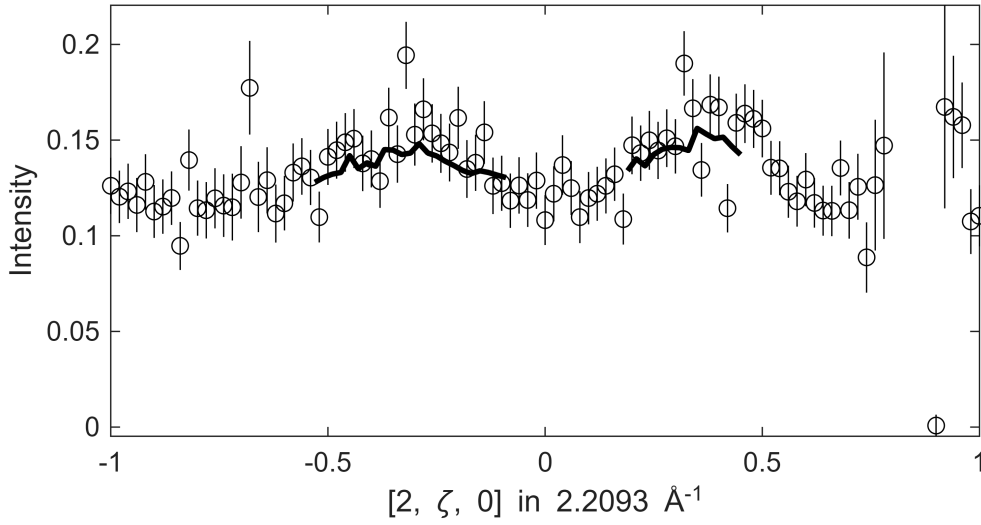


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*****
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*****
*****
*** step: 8#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Fe_ei800, w2_800off200 Sym fits.sqw

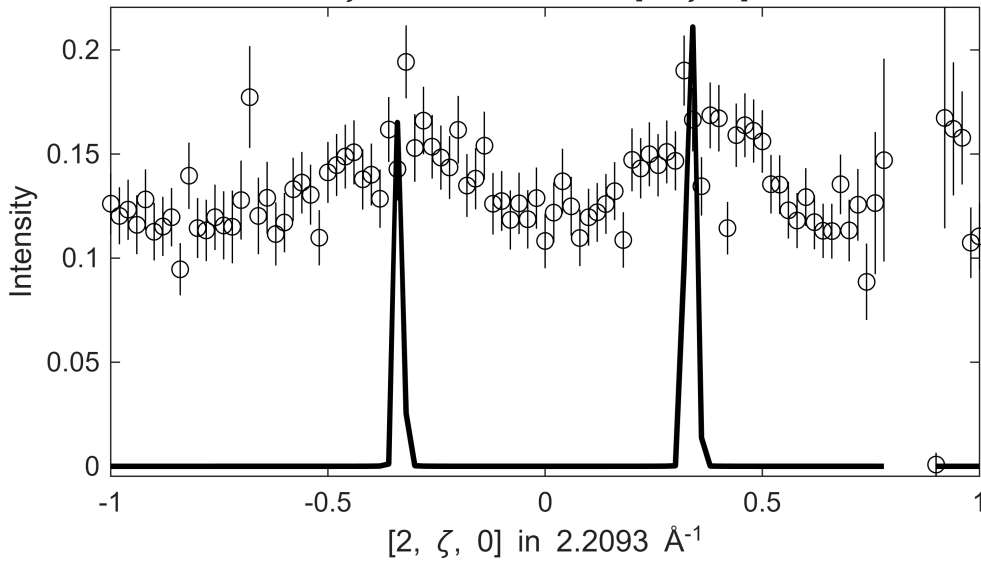
S=0.14±0.018; J0= 36±0.0093; gamma=-0.0974876±0.0280349

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 140 ≤ E ≤ 16
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 140 ≤ E ≤ 16
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

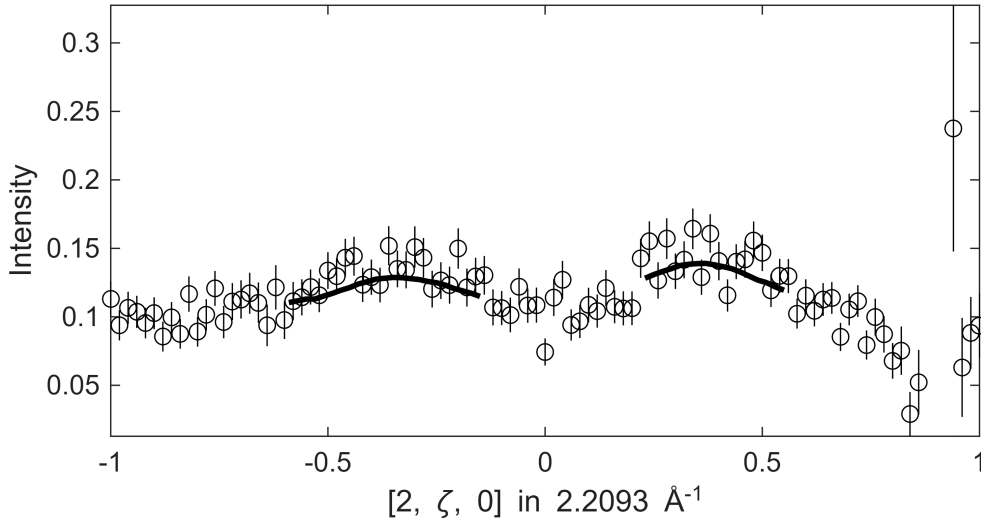


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*****
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*****
*** step: 9#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Fe_ei800, w2_800off200 Sym fits.sqw

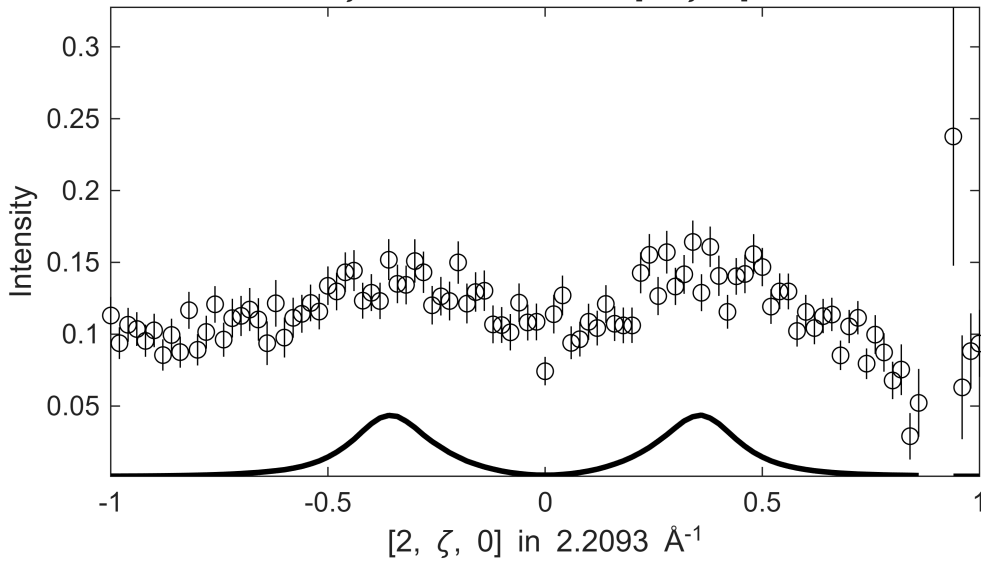
S=0.55±0.041; J0= 41± 6.3; gamma=123.585±40.5543

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 150 ≤ E ≤ 17
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 150 ≤ E ≤ 17
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

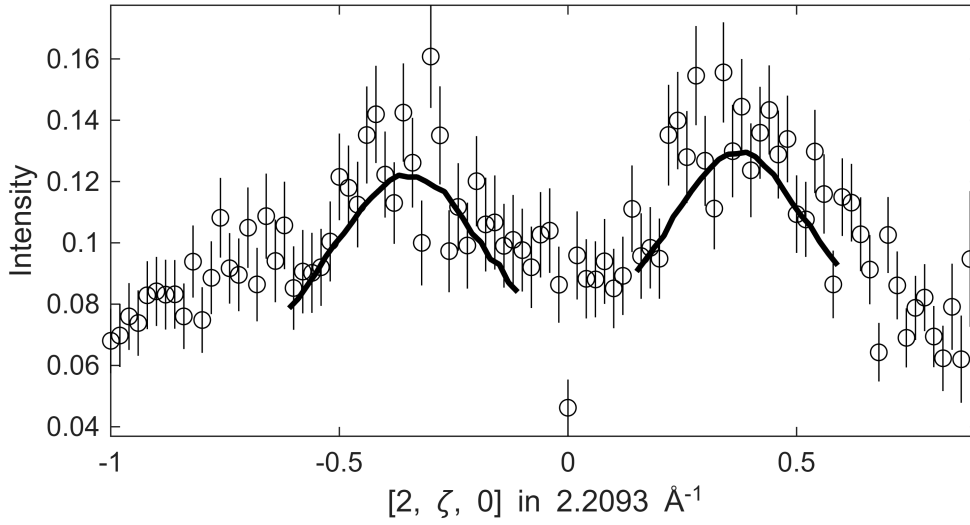


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*****
***  <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<  ***
*****
*** step: 10#25  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>  ***
*****
```

Fe_ei800, w2_800off200 Sym fits.sqw

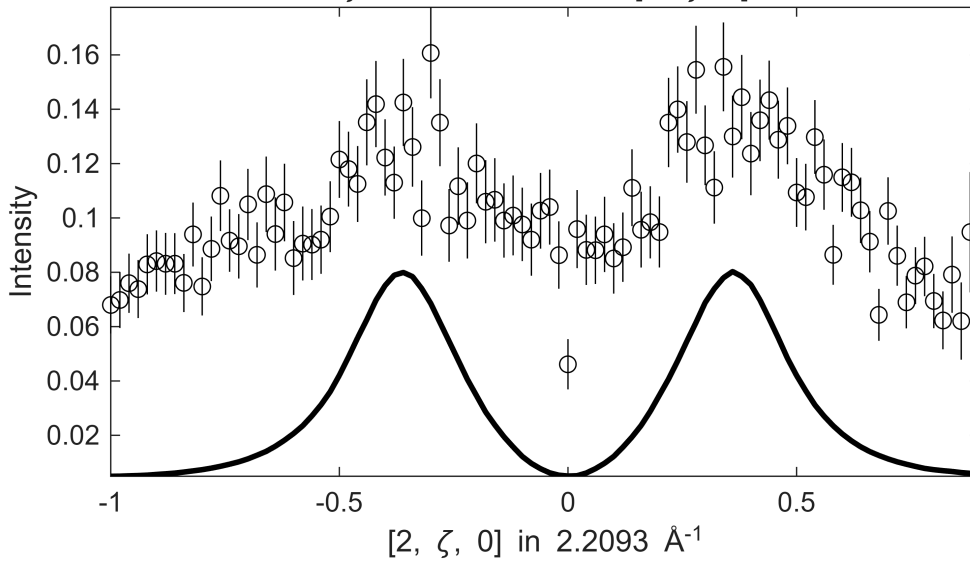
S= 2.3± 4.2; J0= 60± 67; gamma=368.703±946.732

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 160 ≤ E ≤ 18
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 160 ≤ E ≤ 18
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 11#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

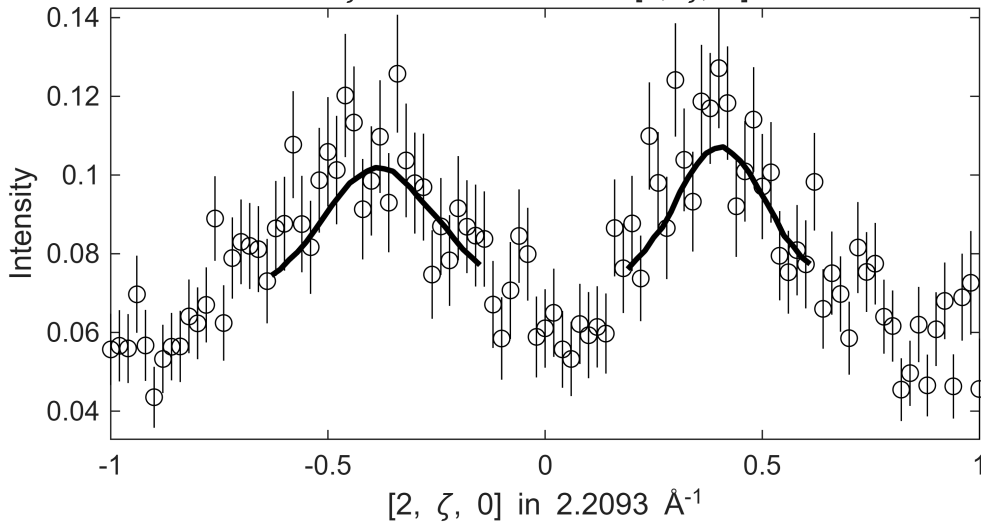
S=0.64±0.78; J0= 39± 11; gamma=102.442±144.459

17

Fe_ei800, w2_800off200 Sym fits.sqw

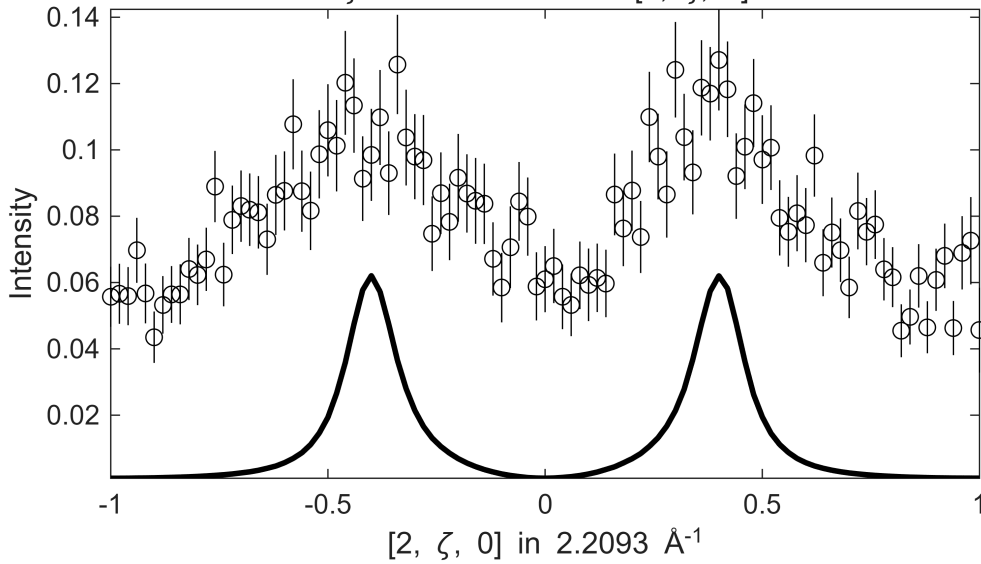
S=0.42±0.35; J0= 35± 3.2; gamma=59.2203±66.7778

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 180 ≤ E ≤ 20
 ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 180 ≤ E ≤ 20
 ζ=-1.01:0.02:1.01 in [2, ζ, 0]

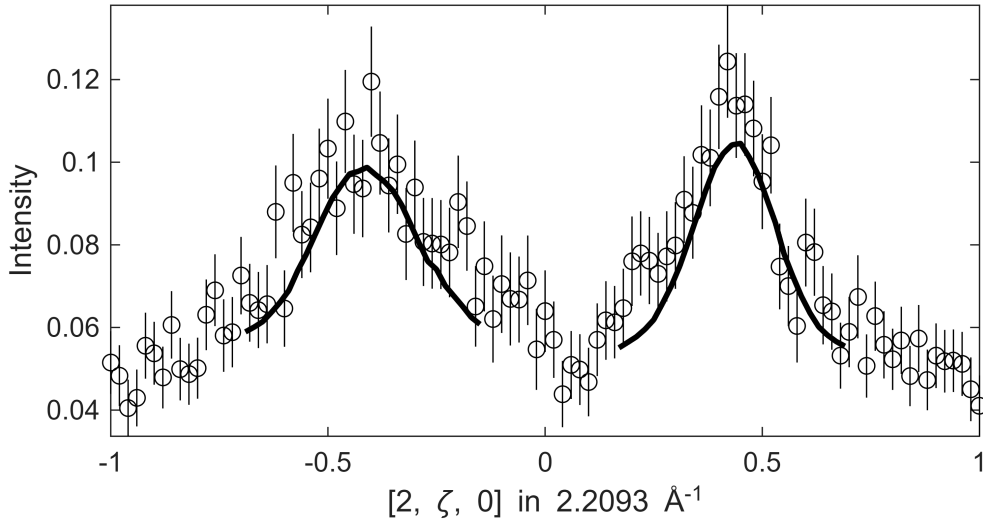


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*****
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*****
*** step: 13#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Fe_ei800, w2_800off200 Sym fits.sqw

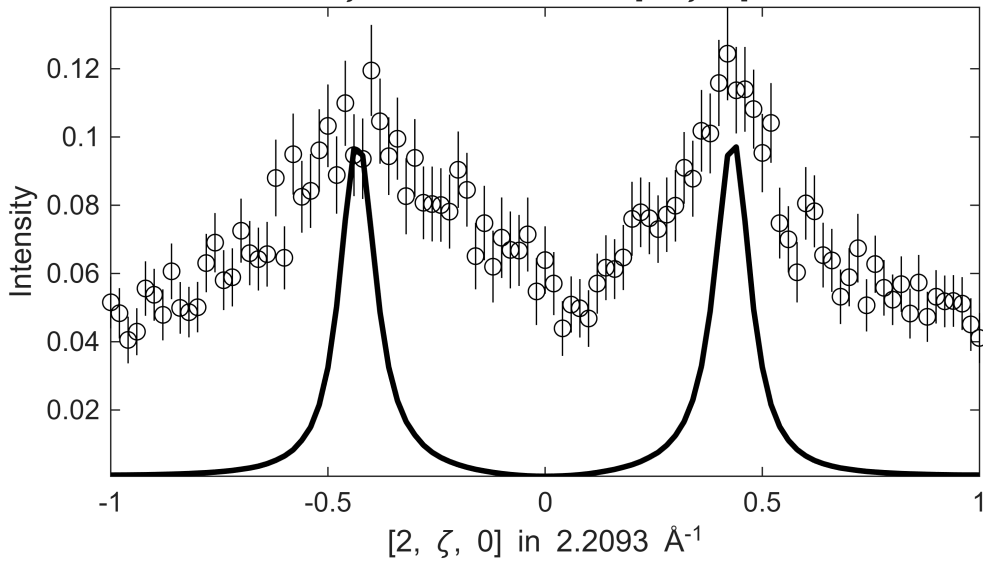
S=0.42±0.018; J0= 32±0.84; gamma=35.6287±3.96633

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 190 ≤ E ≤ 21
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



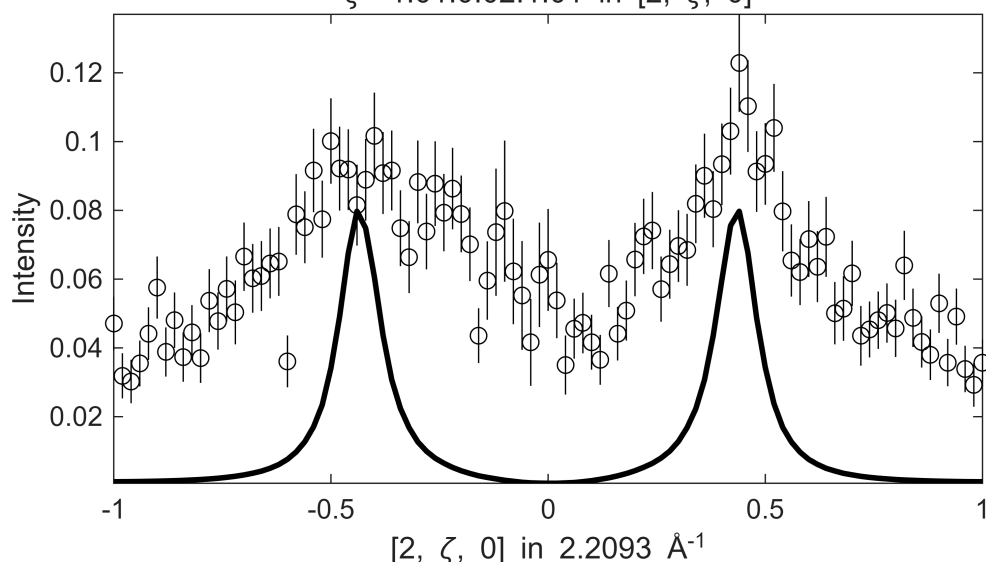
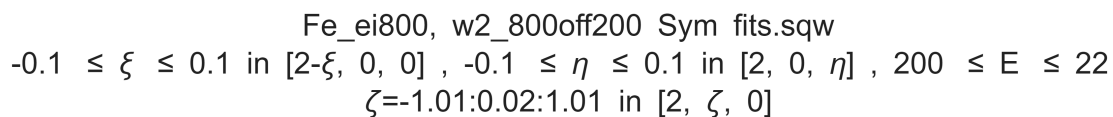
Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 190 ≤ E ≤ 21
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 14#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $200 \leq E \leq 22$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$

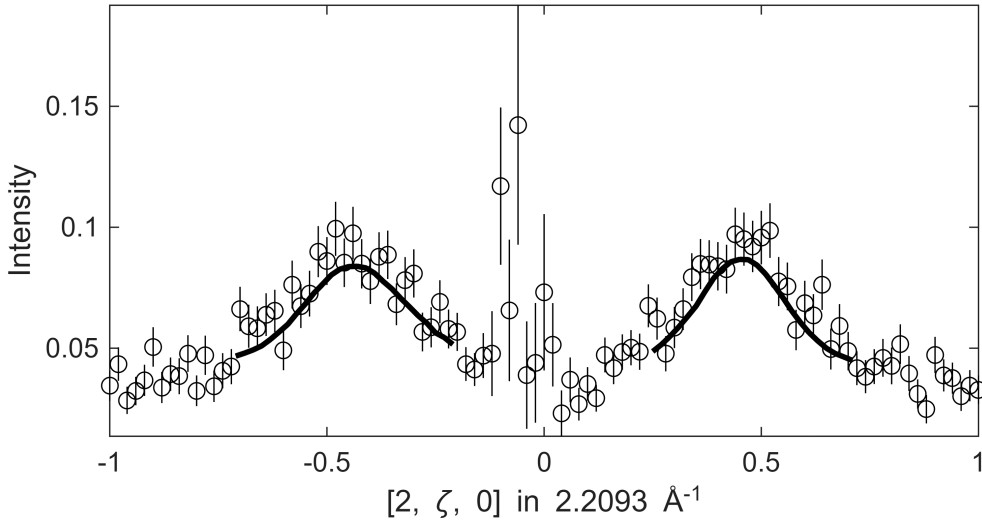


20

Fe_ei800, w2_800off200 Sym fits.sqw

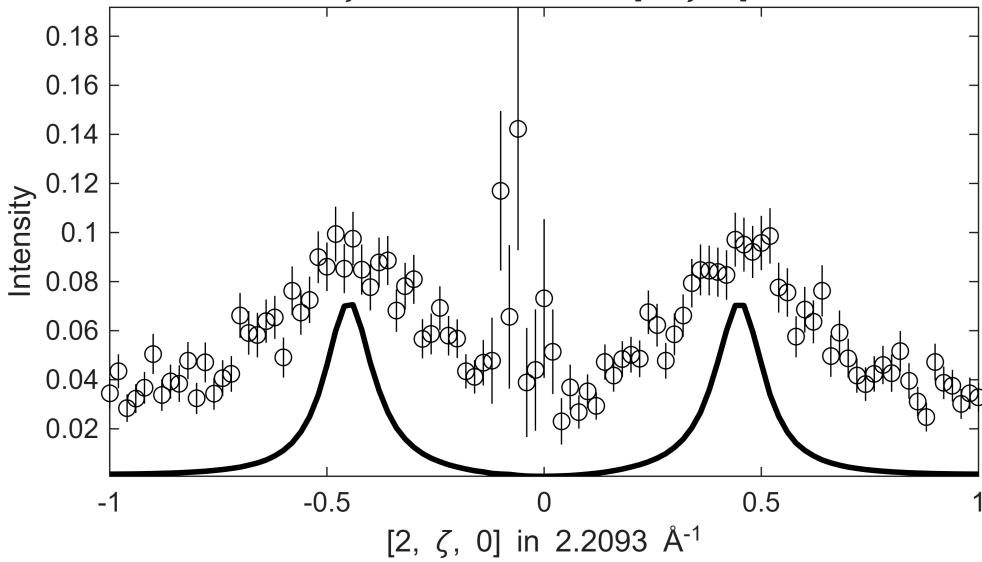
S=0.46±0.21; J0= 33± 1.3; gamma=54.4827±35.8111

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 210 ≤ E ≤ 23
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 210 ≤ E ≤ 23
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

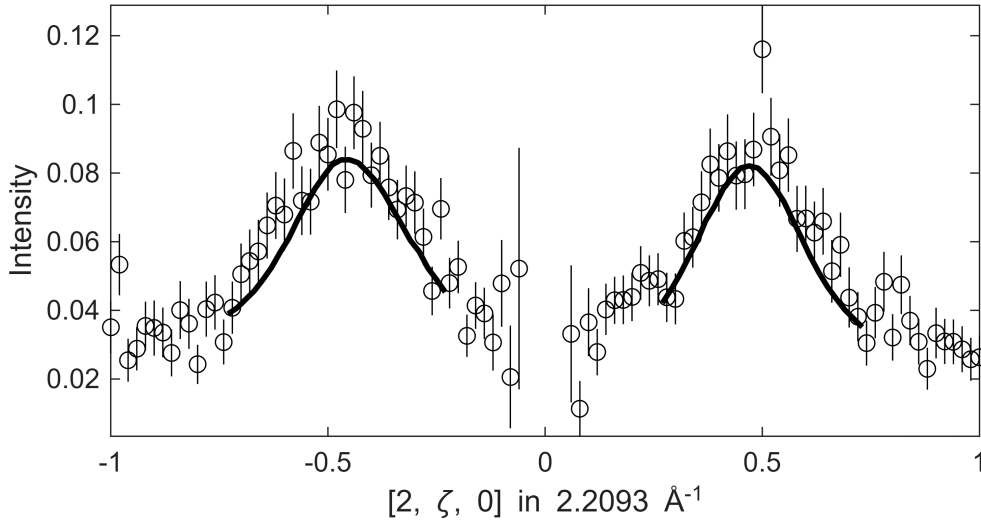


```
*****
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*****
*** step: 16#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Fe_ei800, w2_800off200 Sym fits.sqw

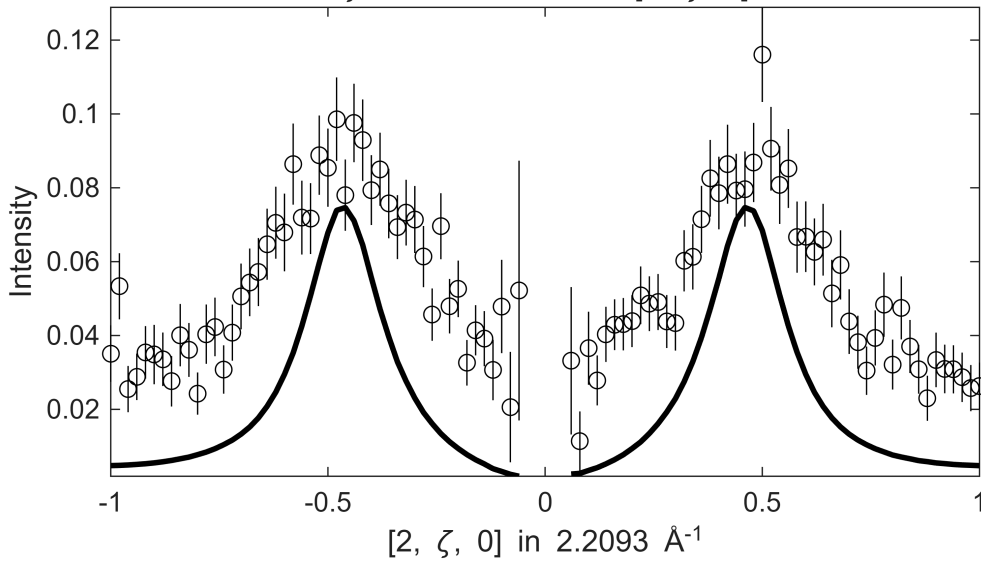
S=0.85± 0.4; J0= 35± 2.4; gamma=100.914±50.0012

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 220 ≤ E ≤ 24
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



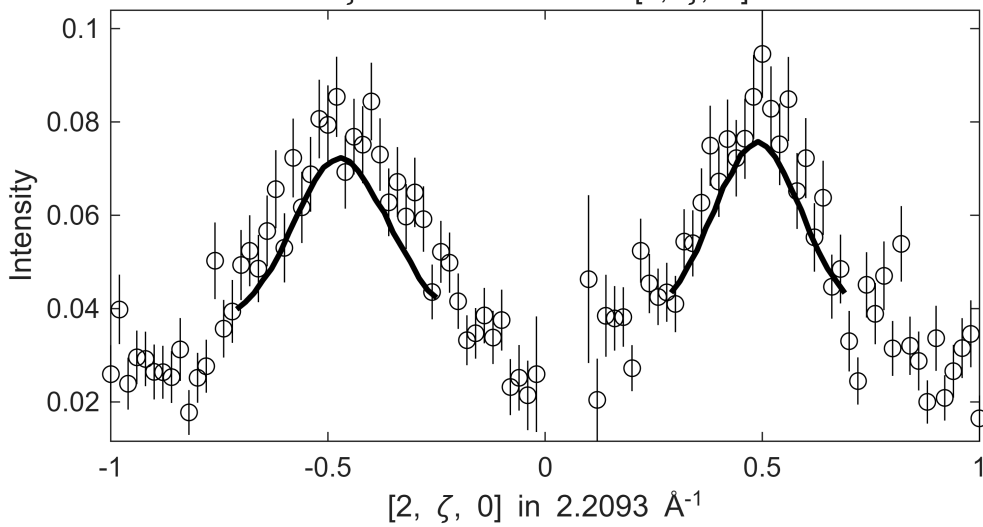
Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 220 ≤ E ≤ 24
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

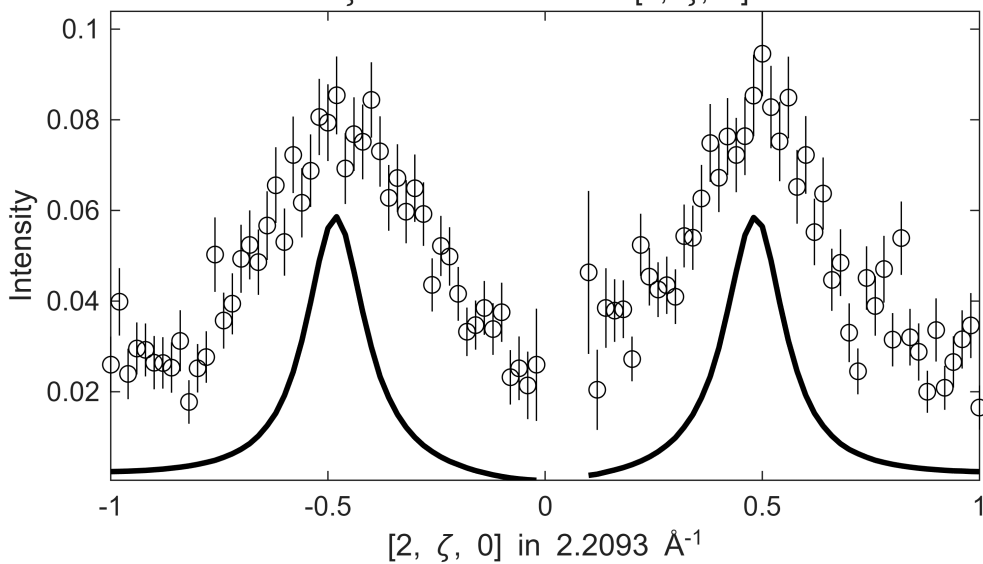


```
*****
***  <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<  ***
*****
*** step: 17#25  >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>  ***
*****
```

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $230 \leq E \leq 25$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$



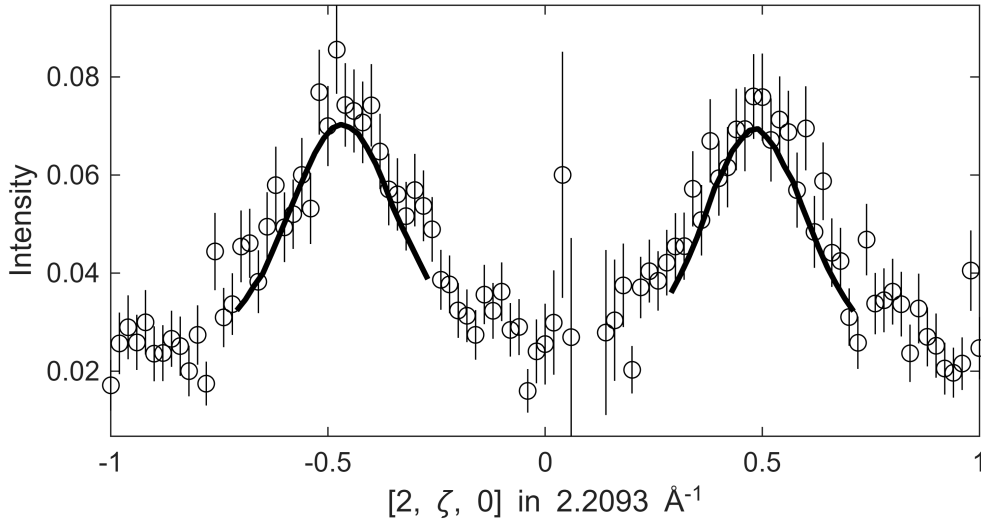
$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $230 \leq E \leq 250$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$

[illegible]

Fe_ei800, w2_800off200 Sym fits.sqw

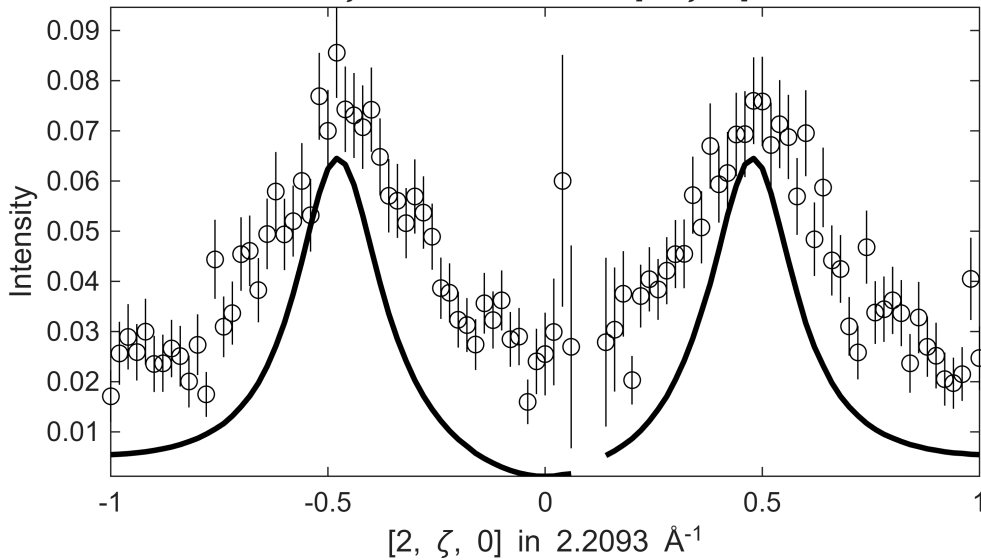
S=0.87±0.56; J0= 37± 3.2; gamma=119.86±70.8087

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 240 ≤ E ≤ 26
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 240 ≤ E ≤ 26
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



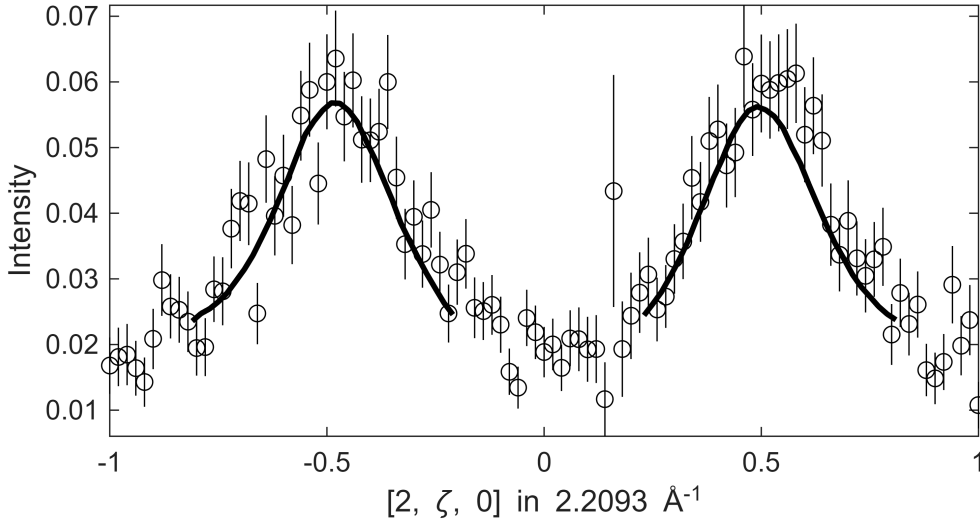
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 19#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.49505
Warning: Dataset 1:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.53763

Fe_ei800, w2_800off200 Sym fits.sqw

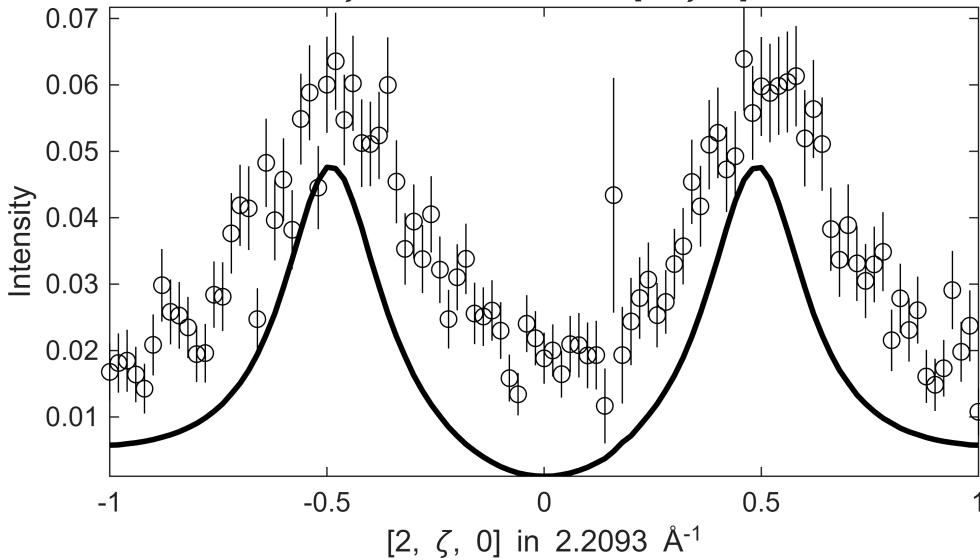
S=0.81±0.27; J0= 39± 2.5; gamma=152.886±51.0433

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 260 ≤ E ≤ 28
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

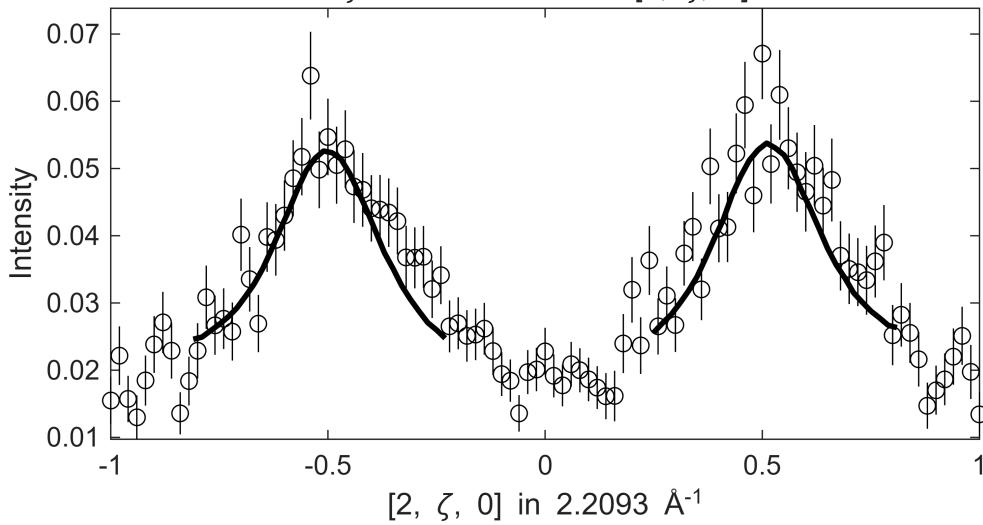
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 260 ≤ E ≤ 28
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



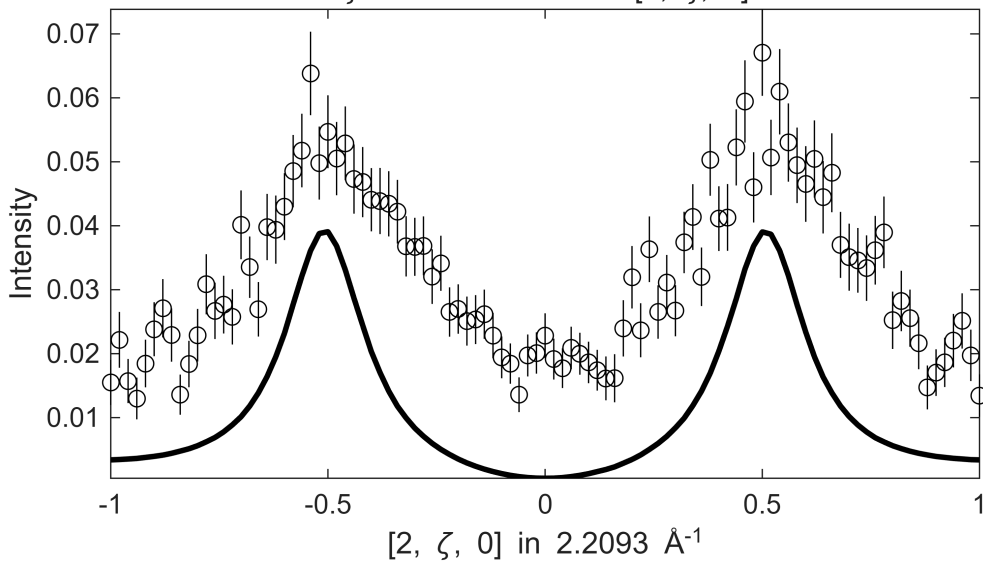
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 20#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.49505
Warning: Dataset 1:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.55556

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 270 ≤ E ≤ 29
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



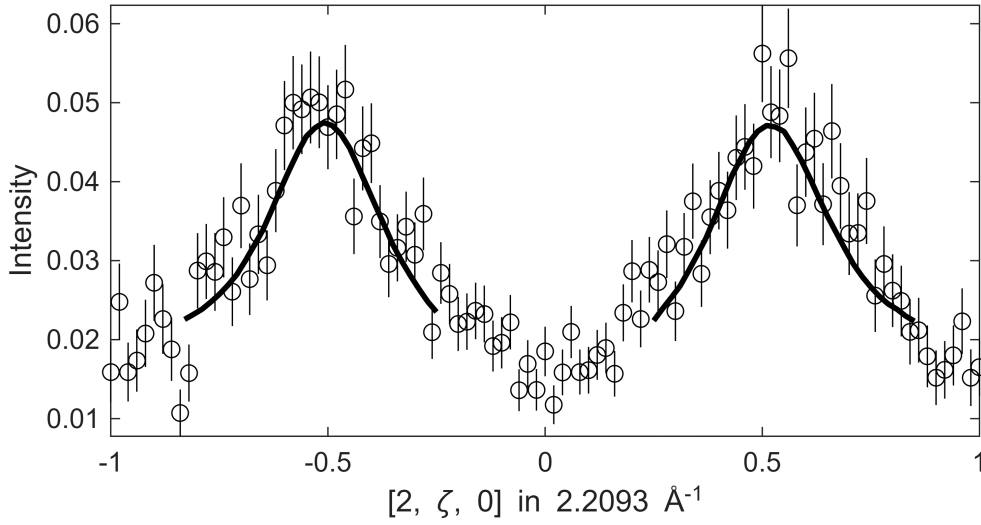
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 270 ≤ E ≤ 29
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

[illegible]

Fe_ei800, w2_800off200 Sym fits.sqw

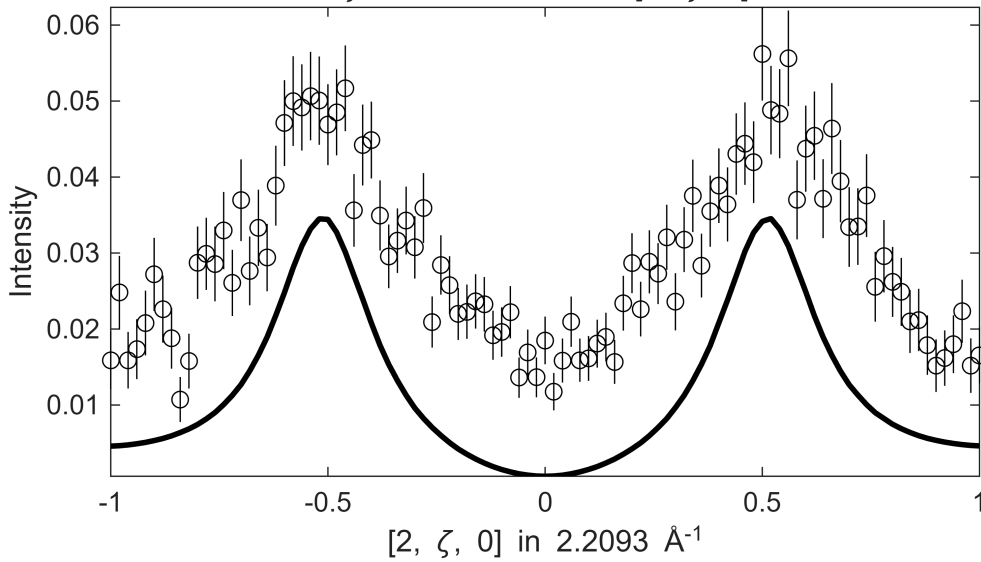
S=0.56±0.18; J0= 38± 1.9; gamma=144.77±43.02

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 280 ≤ E ≤ 30
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 280 ≤ E ≤ 30
ζ=-1.01:0.02:1.01 in [2, ζ, 0]

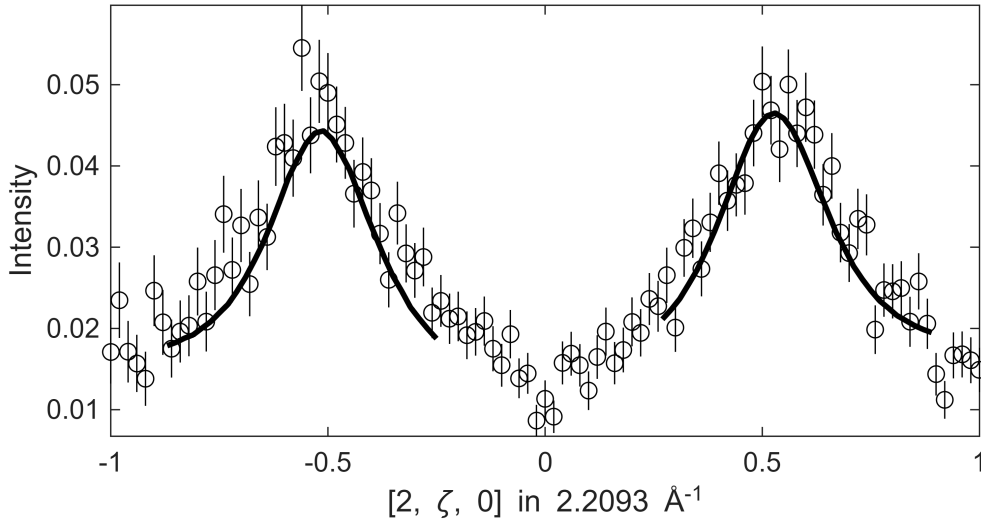


```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*** step: 22#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Fe_ei800, w2_800off200 Sym fits.sqw

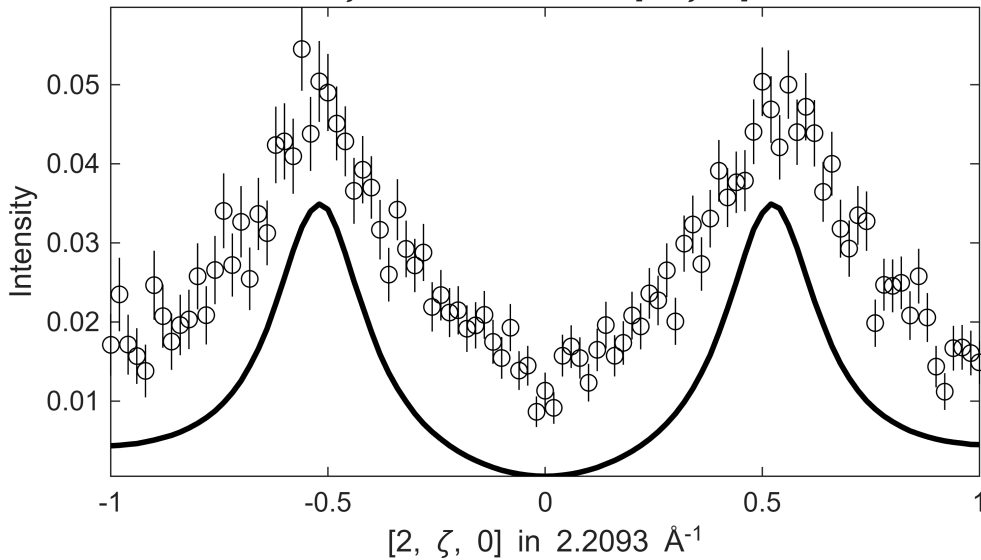
S=0.53±0.0093; J0= 38±0.59; gamma=132.104±5.98489

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 290 ≤ E ≤ 31
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 290 ≤ E ≤ 31
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



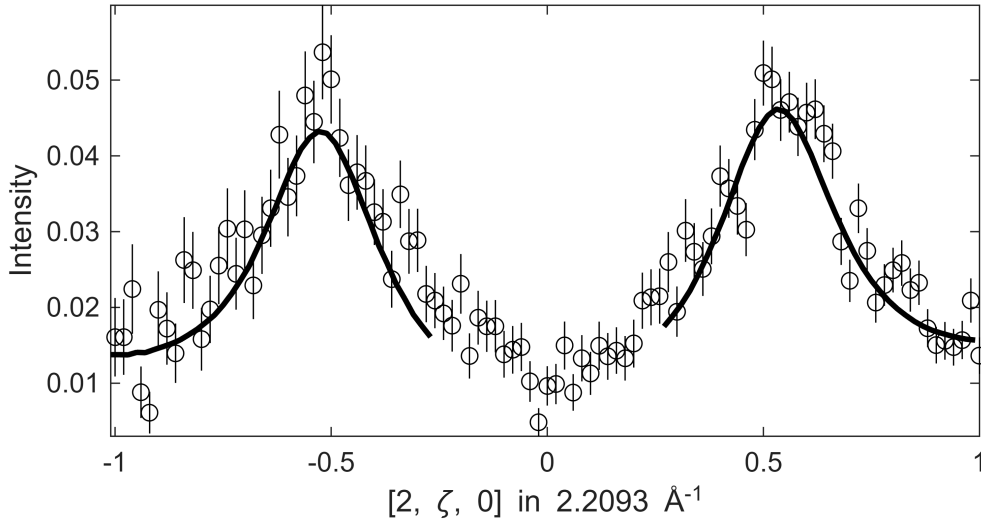
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 23#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.82508
Warning: Dataset 1:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.193

Fe_ei800, w2_800off200 Sym fits.sqw

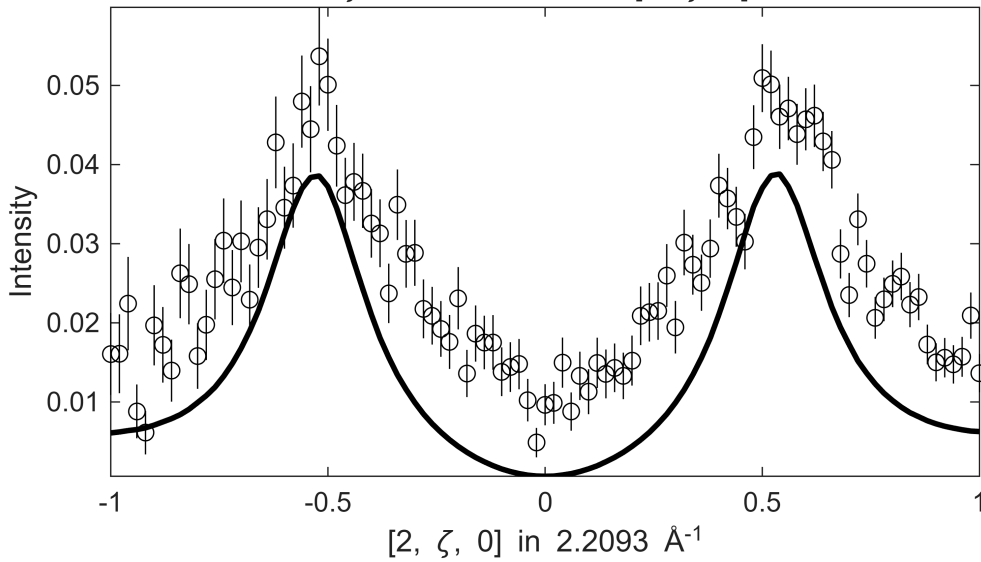
S=0.66±0.12; J0= 39± 1.1; gamma=149.732±25.7596

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 300 ≤ E ≤ 32
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



Fe_ei800, w2_800off200 Sym fits.sqw

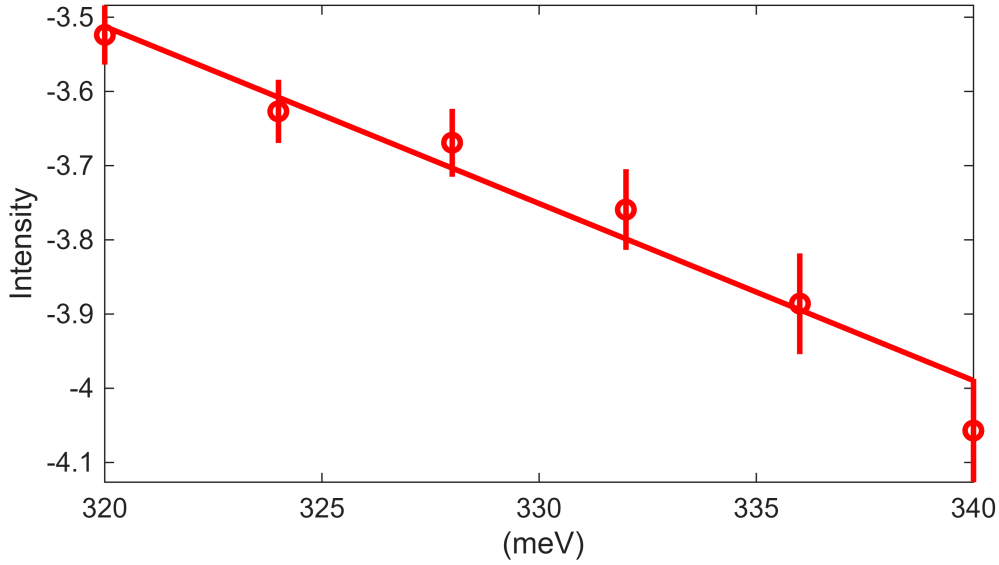
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 300 ≤ E ≤ 32
ζ=-1.01:0.02:1.01 in [2, ζ, 0]



```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*** step: 24#25 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

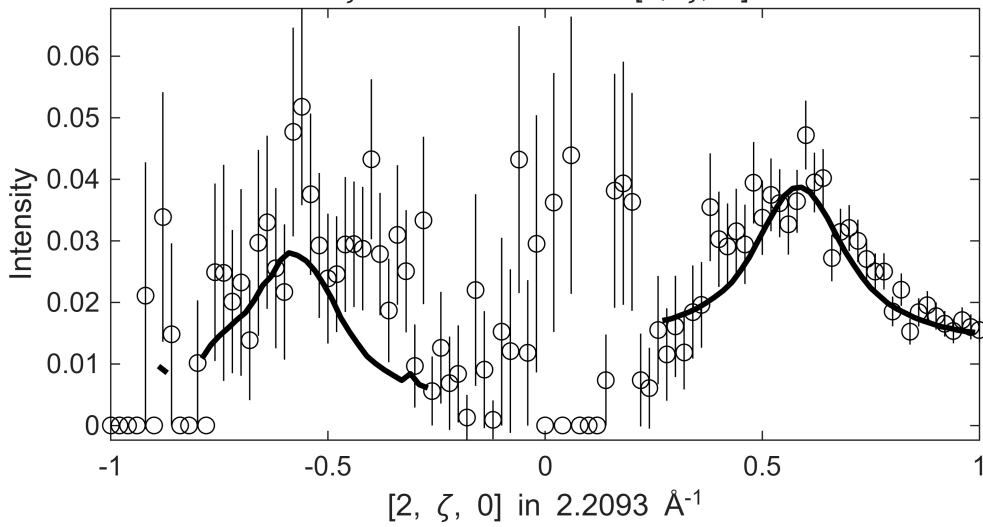
Warning: Dataset:60 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 9.901
Warning: Dataset 1:37 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 15.812

Fe_ei800, w2_800off200 Sym fits.sqw
 $\hat{s} \leq \zeta \leq 1$ in $[2, \zeta, 0]$, $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2$
 $E=318:4:342$



Warning: Dataset 1:25 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 12.2549

Fe_ei800, w2_800off200 Sym fits.sqw
 $S=0.31 \pm 0.01$; $J_0=34 \pm 0.6$; $\gamma=86.2067 \pm 5.14165$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $320 \leq E \leq 34$
 $\zeta=-1.01:0.02:1.01$ in $[2, \zeta, 0]$



Fe_ei800, w2_800off200 Sym fits.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $320 \leq E \leq 34$
 $\zeta = -1.01:0.02:1.01$ in $[2, \zeta, 0]$

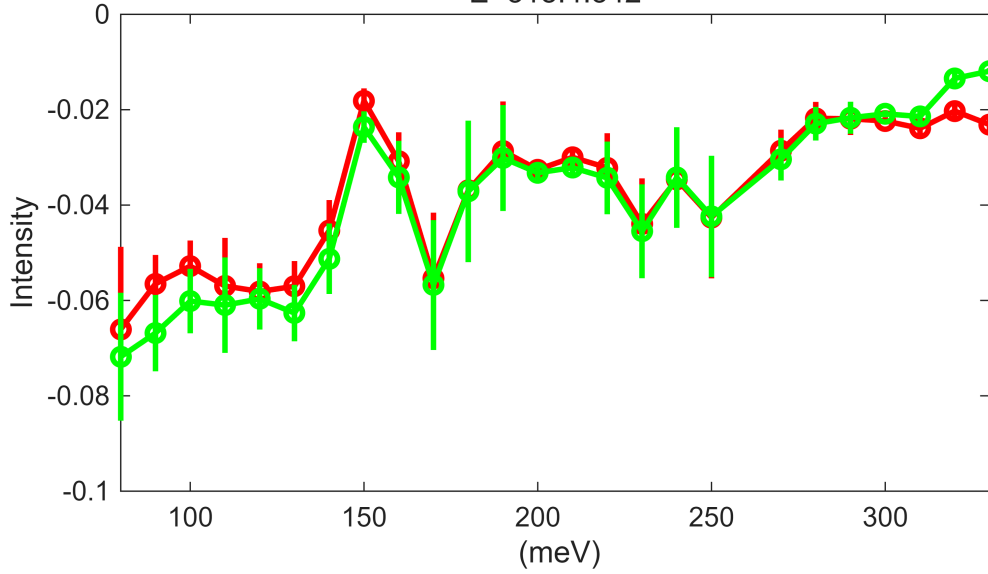


```
gp =  
    genieplot with properties:
```

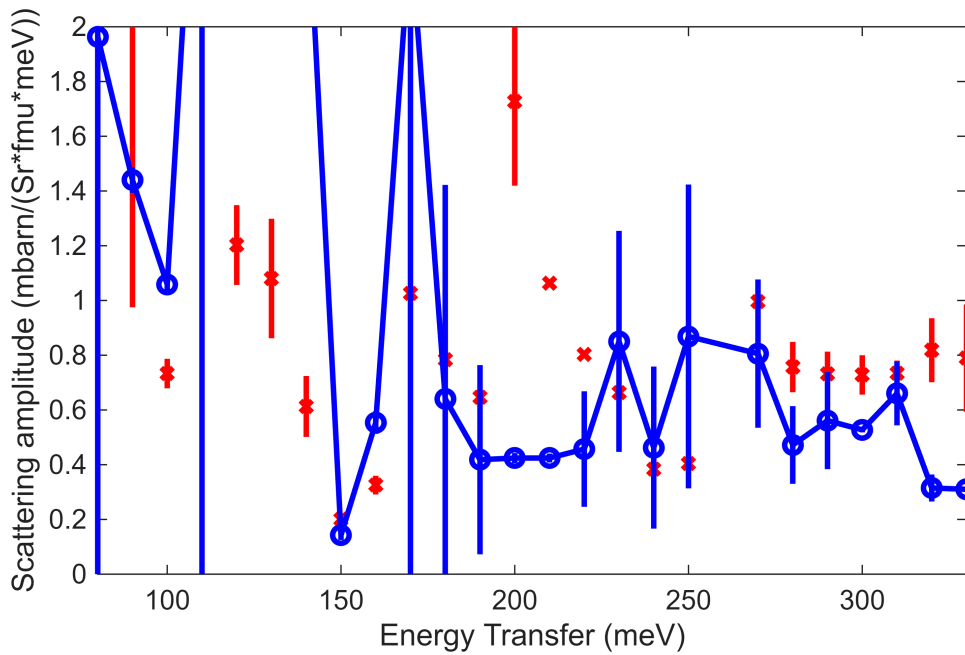
```
use_original_horace_plot_colours: 1
```

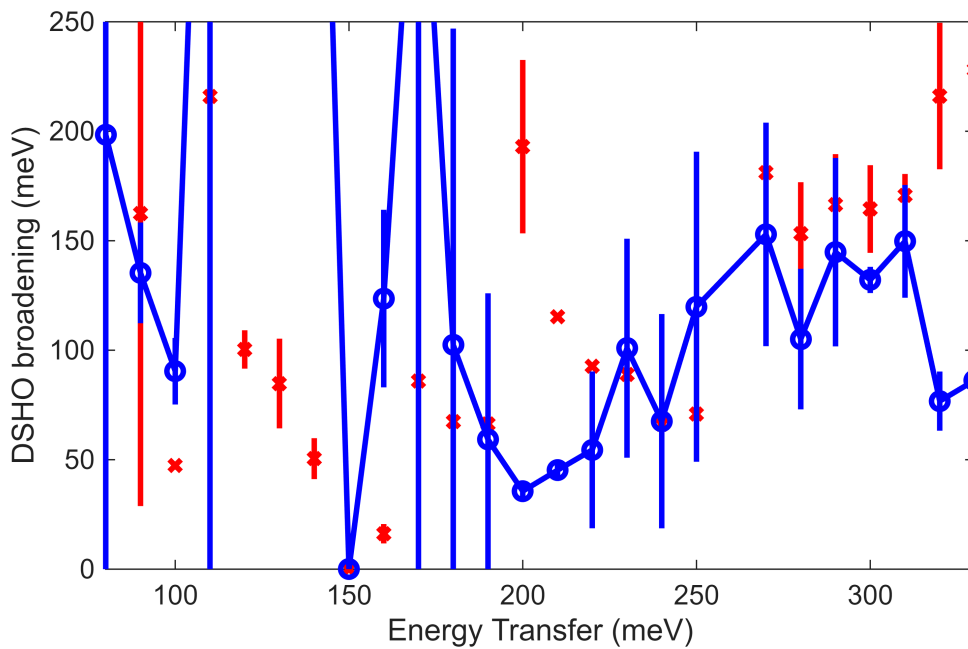
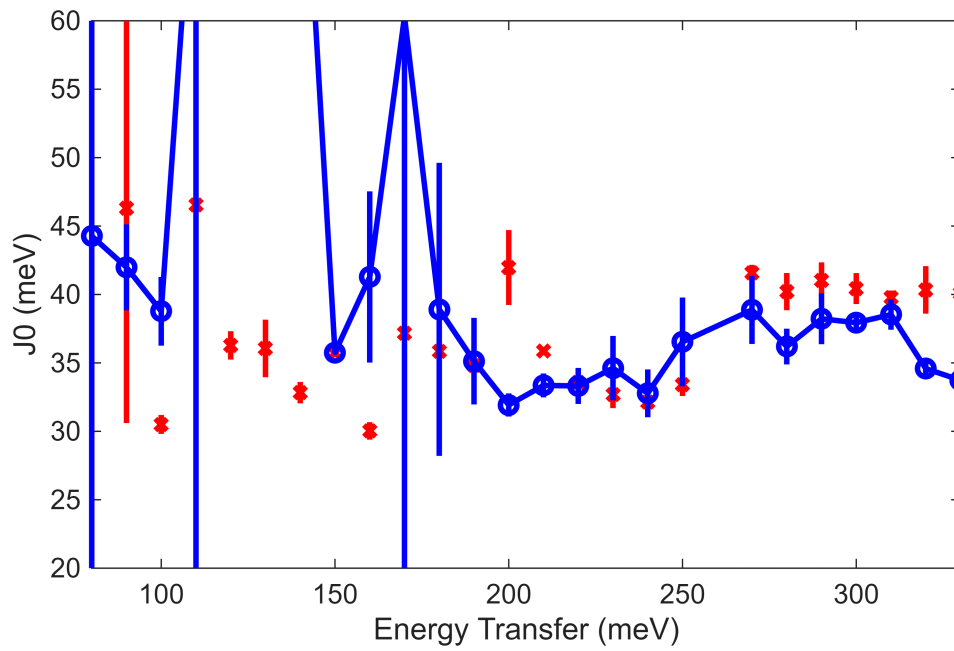
```
pd(bg_data);lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figur
re;ly -0.1 0
```

Fe_ei800, w2_800off200 Sym fits.sqw
 $\zeta \leq 1$ in $[2, \zeta, 0]$, $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2$
 $E=318:4:342$



```
plot_SJG_results(fit_res, 'b', {'S', 'J0', 'gamma'}, {[0,2],[20,60],[0,250]} ,
{Sg,J0,G});
```





```
% fit_res.Speak_varsl = Sg;
% fit_res.J0peak_varsl = J0;
% fit_res.gamma_peak_varsl = G;
fit_res.Speak = Sg;
fit_res.J0peak = J0;
fit_res.gamma_peak = G;

fit_res.peak_fit_locDelSlope_par = fit_res.peak_fit_par;
fit_resEi800 = fit_res;
save(fullfile(fit_data_path,fit_data_file),"fit_resEi800");
```